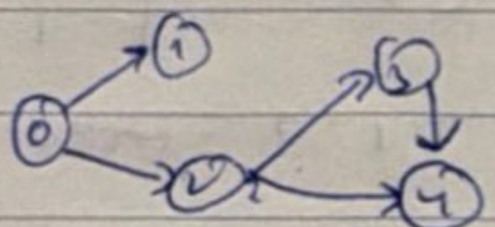
Week-10-11

DAY-6y

1. Depth-First Search:

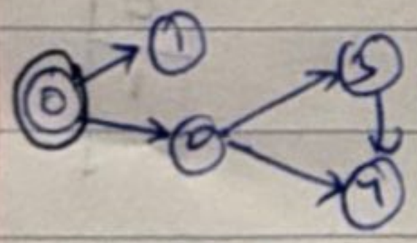
Recursive:

```
def recursivedfs(graph, node, visited=None):
    if visited is None:
        visited = set()
    if node in visited:
        return
    visited.add(node)
    print(node)
    for neighbor in graph[node]:
        recursivedfs(graph, neighbor, visited)
recursivedfs(graph, 0, None)
graph = {0:[1,2], 1:[], 2:[3,4], 3:[], 4:[]}
```



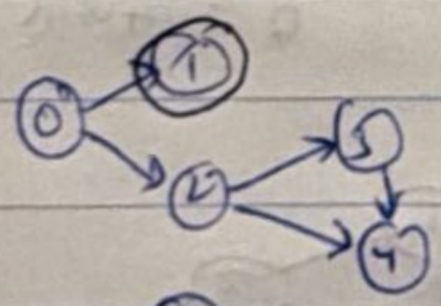
Visited = set()

(i)



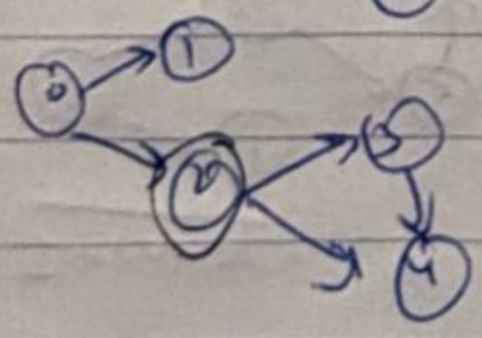
visited = [0]

(ii)



visited = [0,1]

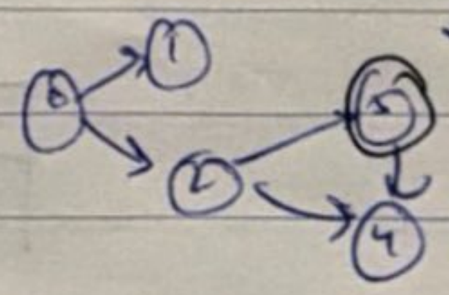
(iii)



visited = 0,1,2

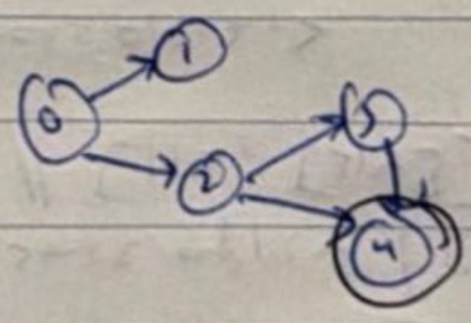
Since 1 has no neighbors.

(iv)



visited = 0,1,2,3

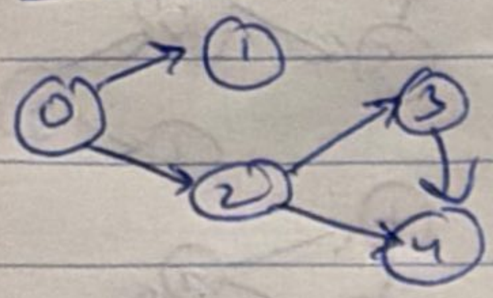
(v)



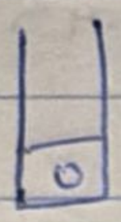
visited = 0,1,2,3,4

Since 3 has no neighbors & 4 has no neighbors

Iterative:-



visited = set()



stack = [1,2]

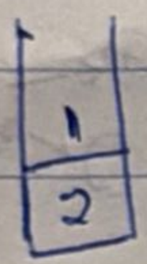
stack.extend ([3,4])

// [1,2,3,4]

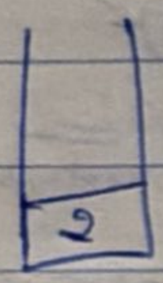
stack.append ([3,4])

// [1,2, [3,4]]

(i) visited = 0

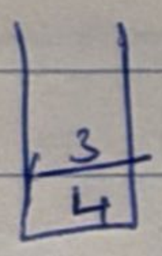


(ii) visited = 0, 1

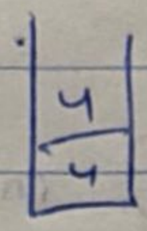


graph[1] = {}

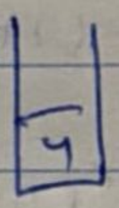
(iii) visited = 0, 1, 2



(iv) visited = 0, 1, 2, 3



(v) visited = 0, 1, 2, 3, 4

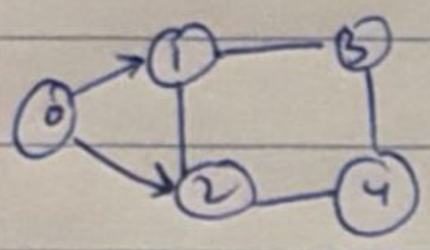


(vi) 4 in visited nothing to do

(vi) stack is empty ← done.

BFS:-

```
def bfs (graph, start):
    visited = set()
    queue = deque([start])
    visited.add(start)
    while queue:
        node = queue.popleft()
        print(node)
        for neigubor in graph[node]:
            if neigubor not in visited:
                visited.add(neigubor)
                queue.append(neigubor)
```



visited = ()

queue = [0]

node = 0 ; visited = 0

(1) visited = 0

que = [1 | 2]

(2) visited = 0, 1, 2

que = [3 | 4]

(3) visited = 0, 1, 2

que = [3 | 4]

(4) visited = 0, 1, 2, 3

que = [4]

(5) visited = 0, 1, 2, 3, 4

que = ()